

HAWK HILLS AGCM, AB, CanadaWMO: **715070**Lat: **57.265N**Lon: **117.298W**Elev: **1493**StdP: **13.92**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.1	-25.6	-37.9	0.7	-31.0	-31.9	1.0	-25.5	23.2	26.2	19.9	16.2	3.3	210	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	83.2	62.5	80.0	61.1	76.9	59.8	65.4	77.6	63.6	75.3	62.0	73.1	8.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	84.2	69.3	59.2	79.3	68.3	57.4	74.4	66.4	31.0	78.4	29.7	75.6	28.5	73.1	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.1	14.7	DB	-38.2	88.9	4.8	2.7	-41.6	90.9	-44.5	92.5	-47.2	94.0	-50.7	95.9	(4)
(5)			WB	-38.3	68.8	4.8	2.3	-41.8	70.5	-44.6	71.9	-47.3	73.2	-50.9	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.7	5.5	6.2	17.5	36.1	50.6	58.6	62.1	59.5	50.1	36.7	15.9	3.9		
	DBStd	24.48	16.43	14.85	14.99	10.19	8.30	5.89	4.71	5.79	8.02	8.83	13.21	15.12		
	HDD50	7113	1379	1227	1008	424	93	8	1	5	94	421	1024	1429		
	HDD65	11477	1844	1647	1473	866	449	202	114	185	450	879	1474	1894		
	CDD50	1161	0	0	0	8	111	266	375	298	96	7	0	0		
	CDD65	52	0	0	0	0	2	11	24	13	2	0	0	0		
	CDH74	1360	0	0	0	11	126	271	531	331	89	1	0	0		
	CDH80	278	0	0	0	1	23	55	114	69	16	0	0	0		
	WSAvg	6.4	6.9	6.3	6.4	7.4	7.8	6.4	5.5	5.4	6.0	6.4	6.2	6.5		
	PrecAvg	19.1	1.1	0.7	0.8	0.9	1.8	3.5	3.3	2.2	1.6	1.1	1.1	0.9		
(15) Precipitation	PrecMax	25.6	2.3	1.2	1.7	1.8	3.5	6.2	5.8	4.2	4.2	2.6	2.2	2.3		
	PrecMin	12.5	0.3	0.2	0.2	0.2	0.6	1.2	1.9	0.7	0.2	0.5	0.4	0.3		
	PrecStd	3.1	0.5	0.3	0.3	0.4	0.7	1.3	1.1	0.8	0.7	0.5	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	42.8	51.1	74.5	83.1	85.8	88.4	86.6	82.3	68.8	48.3	42.3		
		MCWB	38.9	36.4	40.8	52.8	57.5	62.4	65.9	64.9	61.8	53.1	41.8	35.8		
	2%	DB	40.7	38.6	46.3	63.6	77.4	80.5	83.0	81.4	76.0	59.4	42.2	36.0		
		MCWB	35.8	34.0	38.2	46.8	54.5	60.9	63.6	62.7	58.6	47.1	37.2	32.0		
	5%	DB	36.7	33.6	42.4	56.8	72.8	76.8	79.7	77.6	70.1	54.8	38.0	30.8		
		MCWB	33.3	31.0	36.0	42.9	52.2	59.2	62.3	61.3	55.4	45.2	34.0	28.3		
	10%	DB	31.4	27.7	38.4	52.6	69.2	73.2	76.5	74.0	65.2	50.2	33.4	25.3		
		MCWB	28.6	25.4	33.3	41.5	50.8	57.4	61.2	60.2	53.2	42.8	31.0	23.7		
	0.4%	WB	38.6	36.9	41.6	53.1	59.0	66.0	68.8	67.5	63.0	54.3	42.0	36.5		
		MCDB	44.7	41.4	50.1	72.9	79.8	78.5	82.7	80.8	79.0	65.3	47.9	41.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.3	34.2	38.8	47.5	55.9	63.1	65.8	64.8	59.6	48.7	37.4	31.8		
		MCDB	40.4	37.8	45.5	61.1	72.7	75.2	78.0	76.1	72.1	57.5	41.6	35.6		
	5%	WB	33.4	30.6	36.2	44.2	53.8	61.1	63.9	63.2	57.2	45.7	34.3	28.2		
		MCDB	36.5	34.2	42.1	55.2	68.8	72.2	75.7	74.1	68.2	53.2	37.7	30.7		
	10%	WB	28.7	25.3	33.5	41.8	51.9	59.2	62.3	61.5	54.4	43.1	31.0	23.3		
		MCDB	31.6	27.6	38.1	51.4	66.4	69.6	73.5	71.5	63.6	49.4	33.2	24.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	20.4	23.0	22.1	27.0	23.3	25.0	25.0	24.7	17.2	15.4	15.8		
		MCDBR	20.9	21.9	24.2	30.3	35.8	31.2	31.6	31.5	32.4	25.6	19.2	19.3		
		MCWBR	17.1	17.8	17.1	16.5	16.8	14.4	14.7	14.9	16.4	15.4	14.7	16.6		
		MCDBR	20.8	21.7	23.4	28.4	31.0	26.3	29.2	28.4	31.2	24.3	18.6	19.1		
	5% WB	MCWBR	17.2	17.9	16.4	16.1	15.1	12.9	14.9	15.1	16.7	15.3	14.8	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.254	0.280	0.333	0.347	0.362	0.366	0.351	0.310	0.287	0.240	0.216		
	taud		2.283	2.316	2.345	2.362	2.415	2.392	2.403	2.426	2.555	2.569	2.405	2.189		
	Ebn at Noon		219	260	282	280	282	278	274	271	270	249	215	184		
	Edh at Noon		14	22	28	33	33	35	34	31	23	17	13	12		
(44) All-Sky Solar Radiation	RadAvg		199	499	1002	1498	1781	1868	1790	1464	1000	518	216	128		
	RadStd		17	41	75	85	139	161	107	122	73	69	19	10		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (15 neighbors)	N/A	N/A	+3.49	N/A	N/A	N/A	N/A	+90	N/A	(47)

Nomenclature: See separate page